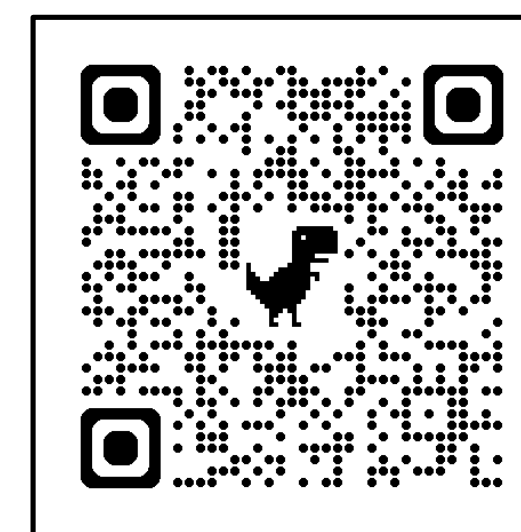


PaP-NF: Probabilistic Long-Term Time Series Forecasting via Prefix-as-Prompt Reprogramming and Normalizing Flows

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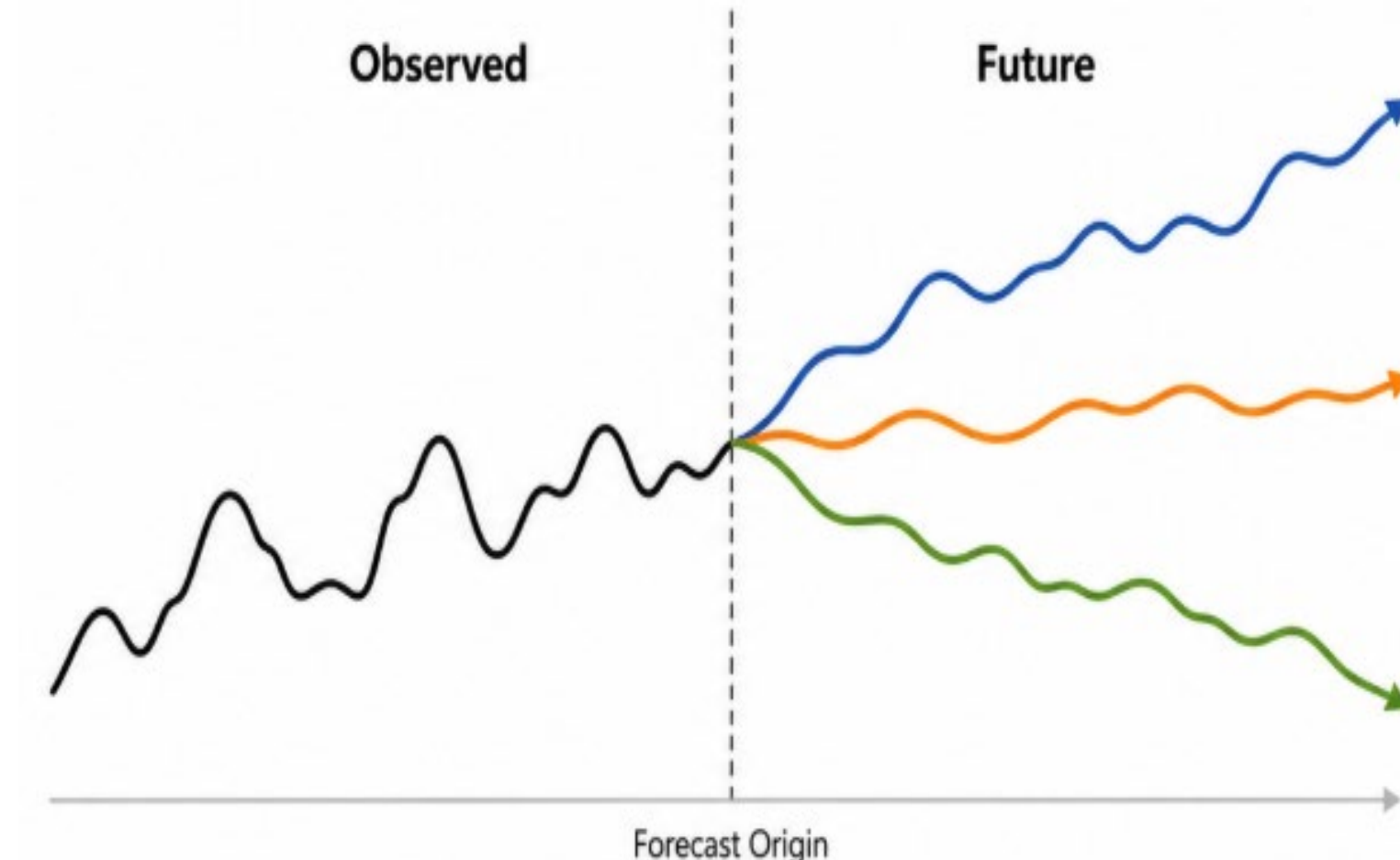


Paper

1. INTRODUCTION

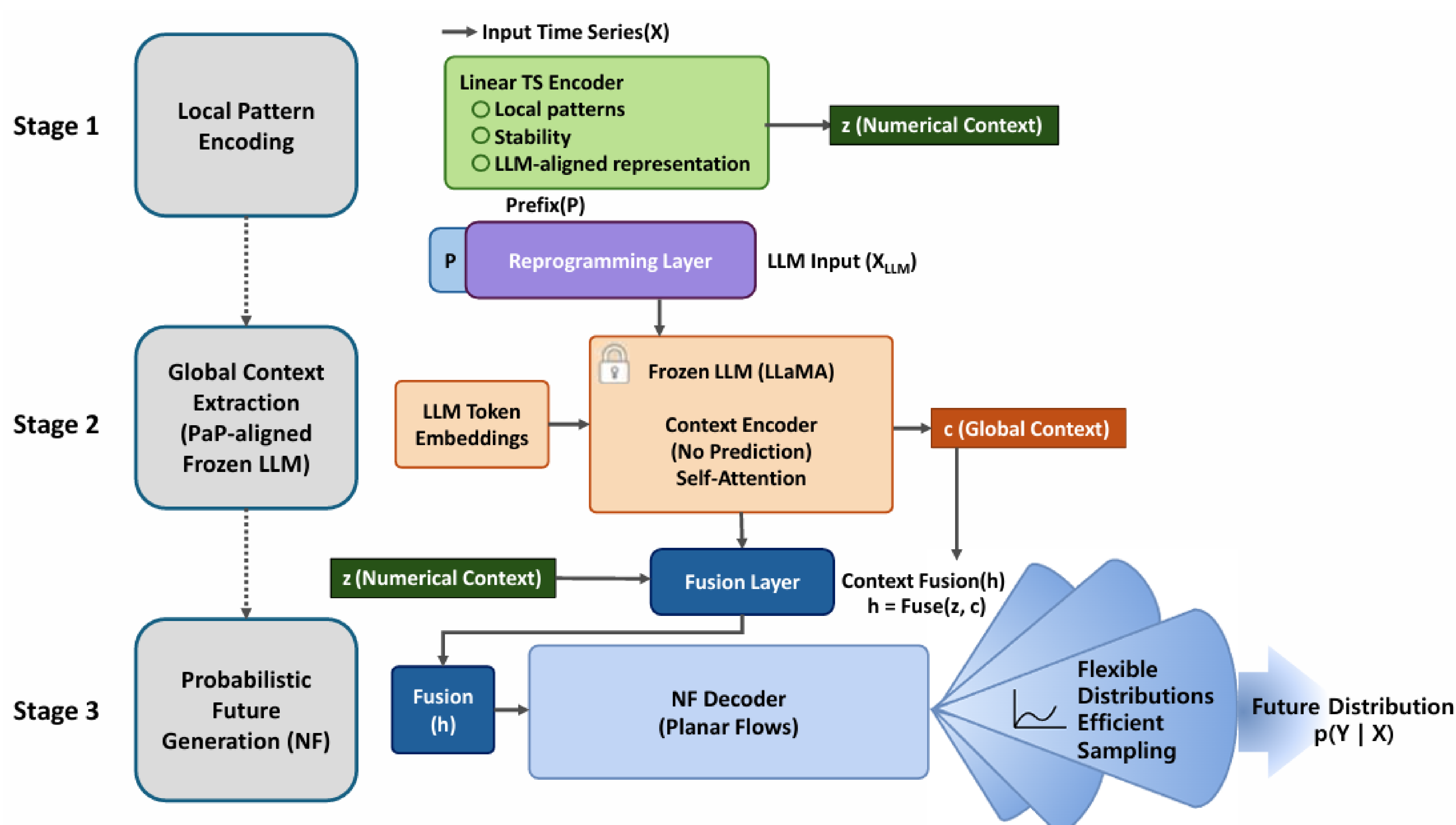
- Long-term time series forecasting is important in energy, healthcare, and traffic systems.
- Longer forecasting horizons increase future uncertainty.
- LLMs capture global context, but struggle to directly predict continuous values.

Can an LLM provide global context without directly predicting values, while modeling uncertainty?



2. OVERVIEW OF PaP-NF

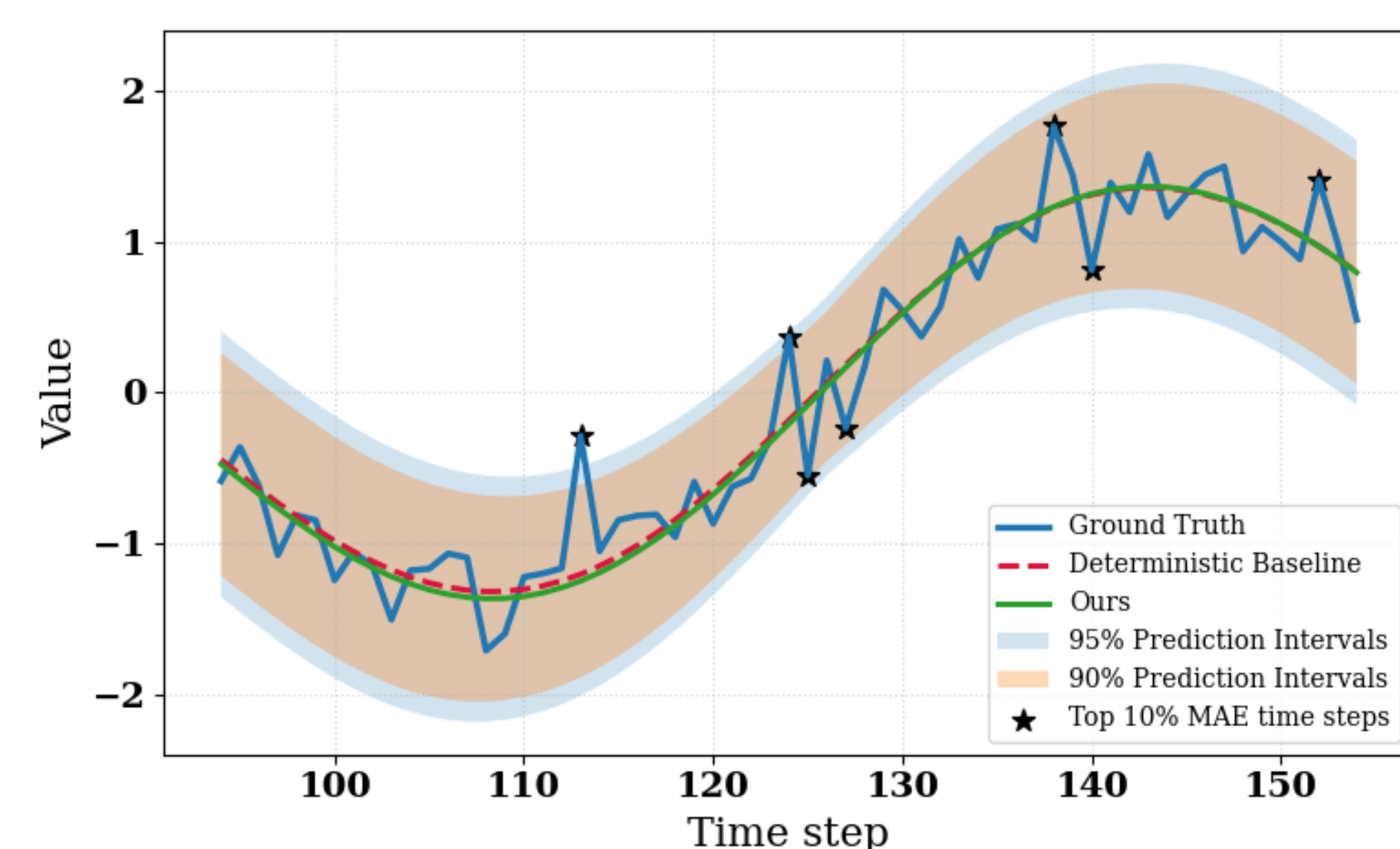
PaP-NF consists of three stages: Local encoding
→ Global context extraction → Probabilistic generation



3. KEY CONTRIBUTIONS

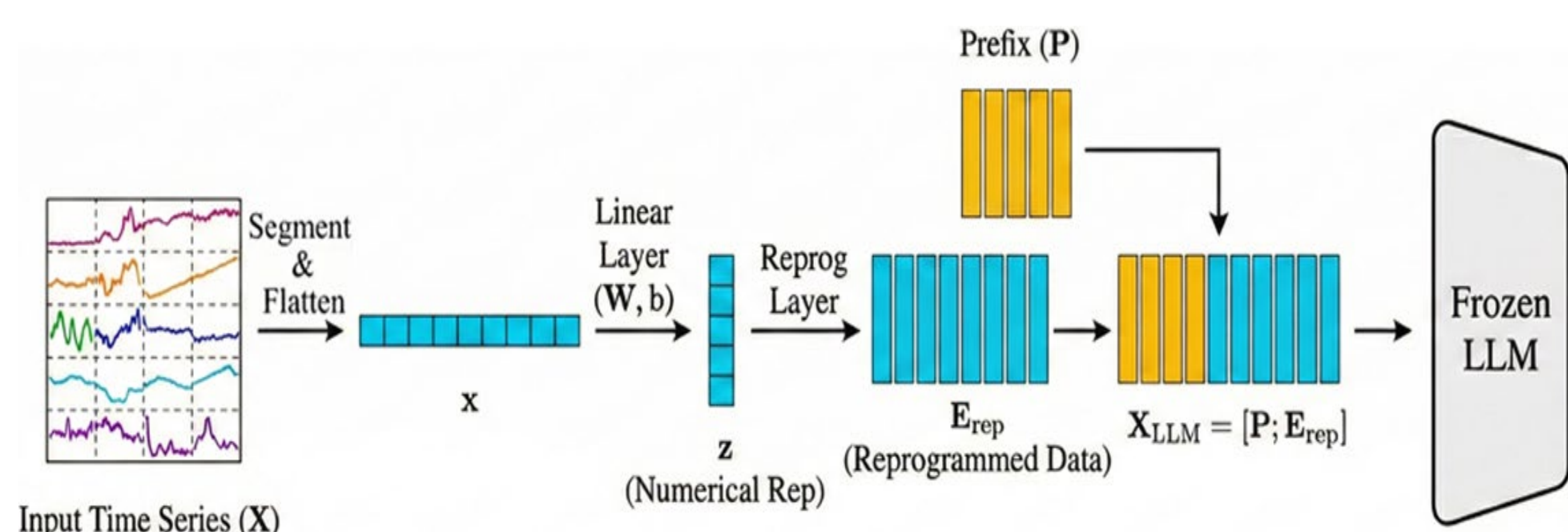
- Hybrid framework:** a linear encoder for local patterns, a frozen LLM for global context and Normalizing Flows for uncertainty.
- Prefix-as-Prompt reprogramming** aligns time series with the LLM embedding space without tuning the LLM.
- Conditional normalizing flows** generate flexible, multimodal future distributions.

4. UNCERTAINTY COVERAGE



- Most challenging points are covered by the 90% prediction intervals.

5. PREFIX-AS-PROMPT ALIGNMENT



- Project numerical features into the LLM token-embedding space.
- Add learnable prefix vectors P .
- PaP bridges time-series representations and a frozen LLM.
- Prefix vectors guide the frozen LLM to interpret reprogrammed time-series features.
- The frozen LLM extracts global context from the PaP-aligned input.

6. EXPERIMENTAL RESULTS

Point Forecasting Performance (MSE / MAE)

Dataset	Model	Prediction Horizon (H)			
		96	192	336	720
ETTh1	FEDformer	0.376/0.419	0.420/0.448	0.459/0.465	0.506/0.507
	TimesNet	0.384/0.402	0.436/0.429	0.491/0.469	0.521/0.500
	PaP-NF	0.366/0.397	0.420/0.426	0.458/0.455	0.503/0.496
ETTm1	FEDformer	0.379/0.419	0.426/0.441	0.445/0.459	0.543/0.490
	TimesNet	0.338/0.375	0.374/0.387	0.410/0.411	0.478/0.450
	PaP-NF	0.334/0.365	0.382/0.371	0.408/0.411	0.462/0.441
Traffic	FEDformer	0.587/0.366	0.604/0.373	0.621/0.383	0.626/0.382
	TimesNet	0.593/0.321	0.617/0.336	0.629/0.336	0.640/0.350
	PaP-NF	0.579/0.315	0.595/0.305	0.612/0.328	0.618/0.337

Probabilistic Forecasting Performance (CRPS)

Dataset	DynOptTheta	DeepAR	Ours
ETTh1	0.117	0.105	0.103
ETTh2	0.085	0.082	0.082

Ablation Studies (ETTh1)			
Horizon	Model	MSE	MAE
$H = 96$	w/o PaP	0.388	0.435
	PaP-NF (Ours)	0.366 (-5.7%)	0.417 (-4.1%)
$H = 720$	w/o PaP	0.521	0.536
	PaP-NF (Ours)	0.472 (-9.4%)	0.495 (-7.7%)

Model	MSE	MAE
PaP-NF (w/o pre-trained LLM)	0.498	0.487
PaP-NF (pre-trained LLM)	0.459	0.455

7. CONCLUSION

- PaP-NF combines **local numerical encoding**, **frozen-LLM context**, and **normalizing flows** for uncertainty-aware forecasting.
- It achieves **competitive point** and **probabilistic performance** across ETT and Traffic benchmarks.
- Ablation studies confirm the value of **PaP alignment**, **pre-trained knowledge**, and **global context**. Future work will focus on distillation and quantization.